

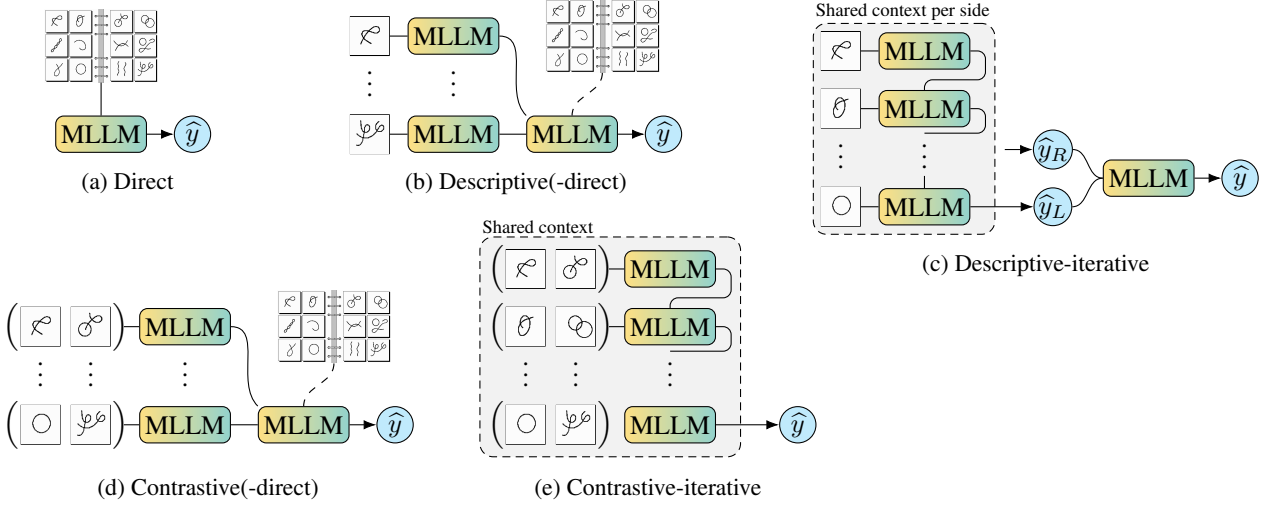


Figure 2: **Generation strategies.** Direct (a) feeds the image of the whole matrix to the model. Descriptive (b), Contrastive (d), and their iterative variants, (c) and (e), present individual image panels to the model in a fixed order. Their direct variants, (b) and (d), additionally include the image of the whole matrix. Grey background marks a sequence of requests run in a single context window.

Descriptive (Fig. 2b). Defines a more granular approach in which the model is first requested to generate a textual description of each image panel of the matrix. Each description is generated in a separate context, such that the model doesn’t have access to the prior panels nor to their descriptions. Next, the model is requested to provide an answer to the problem based only on the generated textual descriptions of all image panels. In this strategy and in all subsequent ones, we evaluate only the final answer provided by the model, without considering the intermediate outputs.

Descriptive-iterative (Fig. 2c). Evaluates the role of the reasoning context and utilizes a context window comprising the dialog history concerning all images in the given side of the problem. After generating the description of the first image, the model iteratively refines its output based on subsequent images from the same side. Based on the textual descriptions of both sides of the problem, the model is requested to provide the final answer.

Descriptive-direct (Fig. 2b with a dashed element). In both above Descriptive strategies, the model is never presented with the image of the whole matrix $\mathcal{BP}^X$. Descriptive-direct strategy extends Descriptive by providing the image of $\mathcal{BP}^X$ along with the textual panel descriptions.

Contrastive (Fig. 2d). A critical aspect of BPs is the focus on forming concepts within the specific context of the matrix $\mathcal{BP}^X$. It’s often the case that correct identification of the concept governing one side requires analysis of the other side to identify their key differences. In Descriptive strategies, the model provides image descriptions concerning a single problem side $\mathcal{L}^X$ or $\mathcal{R}^X$ without taking into

account the images from the other side. Conversely, in the Contrastive strategy, the model is tasked with describing the difference between a pair of corresponding images from both sides of the problem $(L_1^X, R_1^X), \dots, (L_6^X, R_6^X)$. After describing the differences between all six image pairs in separate contexts, the model generates its final answer based on these textual descriptions.

Contrastive-iterative (Fig. 2e). Extends Contrastive by performing all reasoning steps in a single context window, enabling the model to gradually improve its understanding of the rule separating both sides.

Contrastive-direct (Fig. 2d with a dashed element). Extends Contrastive by adding the image of the whole matrix to textual descriptions of differences within each panel pair.

3.2. Evaluation of Natural Language Solutions

The correct answer to a BP may be formulated in natural language in many different ways. To account for this inherent variability, we utilize a model-based approach to assess whether the generated answer $\hat{y}$ matches the ground-truth y . In the proposed setting, an MLLM ensemble receives both $\hat{y}$ and y and is requested to output a binary yes/no answer whether both descriptions refer to the same concept (see Fig. 3a). Specifically, we assess the correctness of $\hat{y}$ using all four considered proprietary MLLMs and count it as correct if at least two models agree with this class (see details in Appendix E). In contrast to solving the BP task, which requires abstract reasoning abilities, the task of determining whether two answers refer to the same concept boils down to assessing the semantic similarity between two texts, and

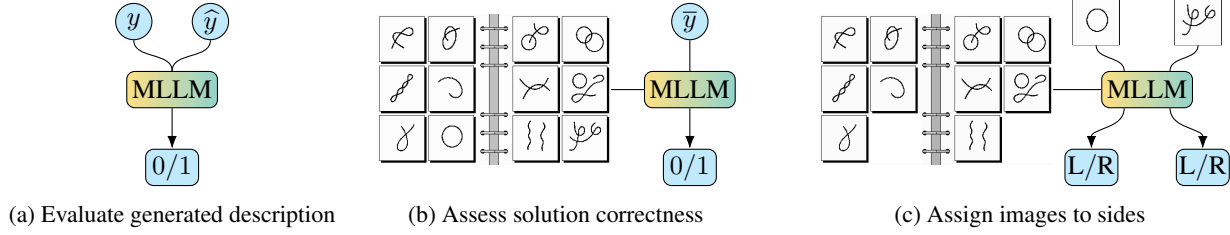


Figure 3: **Evaluation settings.** We consider the three settings to solve BPs: (a) the ground-truth answer y is paired with a description $\hat{y}$ generated by the MLLM and the model needs to verify if they describe the same concepts; (b) given a possible solution $\hat{y}$ the model needs to assess whether it’s correct; (c) two test images corresponding to the left (L) and right (R) sides, resp., are randomly shuffled, and the model needs to assign each image from the pair to the proper side of the problem.

MLLMs are known to excel in such setup (Lu et al., 2023).

3.3. Binary Classification Formulations

To dive deeper into the AVR capabilities of the studied models, we cast the BP task into three binary classification settings, reducing the task’s difficulty.

Firstly, we provide the model with the image of the whole matrix $\mathcal{BP}^X$ along with a possible solution, and the model is prompted to generate a binary score assessing the correctness of the provided answer (Fig. 3b). Two settings are considered, in which the solution is formed by either the actual ground-truth answer (the expected answer is *yes*), or by an incorrect answer taken from a different matrix $\mathcal{BP}^{\mathcal{X}}$, where $\mathcal{X} = (X + 20) \bmod 100$ (the expected answer is *no*). We manually reviewed the resulting pairings to confirm that the selected negative concept was distinct from the original concept. We refer to these two classification setups as Ground-Truth and Incorrect Label, resp.

Secondly, we follow a setting exemplified in the Bongard-LOGO dataset (Nie et al., 2020) in which two test images have to be classified to different sides of the problem (Fig. 3c). To this end, we take two test images corresponding to the respective sides of the matrix, randomly shuffle the images, and request the model to determine the side to which each image belongs. In synthetic BPs we create the test set by removing the 6th image from each side of the matrix, while in BPs from Bongard HOI, Bongard-OpenWorld and Bongard-RWR we use the additional test images. We refer to this setup as Images to Sides.

Prompts for all three binary classification formulations are listed in Appendix I.3.

4. Bongard-RWR

One of the interesting research avenues is to compare the MLLMs performance on synthetic BPs vs. real-world ones. Note, however, that a direct performance comparison on synthetic Bongard dataset vs. real-world Bongard HOI and

Bongard-OpenWorld datasets is not meaningful, as these datasets depict **different concepts**. To enable a meaningful comparison and additionally determine whether the MLLMs performance score is domain-related, we introduce Bongard Real-World Representations, **a focused dataset that expresses concepts present in synthetic BPs using real-world images, thus creating their real-life equivalents**, as illustrated in Fig. 1. Appendix F contains additional examples.

Dataset generation. For a given instance $\mathcal{BP}^X$, we first use GPT-4o to describe its underlying concept y^X in $N = 10$ different ways using the prompt listed in Prompt 4 (Appendix I). We obtain N real-world textual descriptions $D_i^X = \{D_i^{XL}, D_i^{XR}\}$, $i = 0, \dots, N - 1$, of each side $S \in \{L, R\}$. Then, we use image search engine Pexels API (Pexels, 2024) to download $M = 15$ images per each described side D_i^{XS} . We employ GPT-4o (see Prompt 5 in Appendix I) to select only those images that properly illustrate the concept of the respective side and are indeed distinguishable from the alternative concept. We stop the selection procedure after having a set of $T = 3$ descriptions $\{D_{i_1}^X, D_{i_2}^X, D_{i_3}^X\}$, each with 2 appropriate images: $I_k^{XS}(1), I_k^{XS}(2)$, $k = i_1, i_2, i_3$ per each side S (6 left and 6 right ones).

To decrease the possibility of generating a problem with a trivial answer, which is more probable if the images come from a common textual description D_i^X , the corresponding real-world problem instance $\mathcal{RWR}^X = \{\mathcal{L}^X, \mathcal{R}^X\}$ is constructed as follows (see Algorithm 1 in Appendix F): $\mathcal{S}^X = \{I_{i_1}^{XS}(1), I_{i_2}^{XS}(1), I_{i_3}^{XS}(1), I_{i_1}^{XS}(2), I_{i_2}^{XS}(2), I_{i_3}^{XS}(2)\}$, $S \in \{L, R\}$.

We run Algorithm 1 for the first 100 synthetic BPs. After applying the exclusion criteria, this led to the generation of 50 instances $\mathcal{RWR}^X$. However, as we noticed through a manual inspection, some of them were not well depicting the respective problem concept. Hence, we modified the dataset in the following way: 14 problems were entirely removed and, out of the remaining 36, 24 were adjusted through

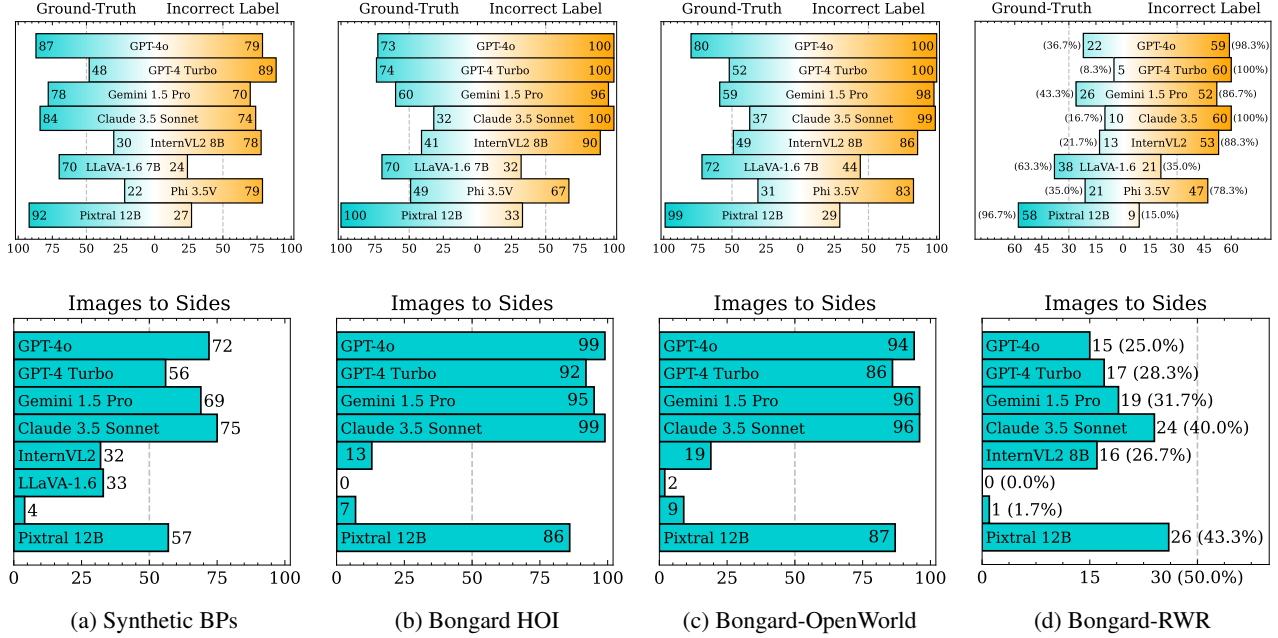


Figure 4: **Binary classification.** The top row presents the results for the Ground-Truth and Incorrect Label strategies, while the bottom row refers to the Images to Sides strategy. Results of a random baseline are marked with a dashed line. For Bongard-RWR, we also report the percentage of correctly solved problems in parentheses.

a manual selection of the images that well represent the considered concept. Furthermore, we extended the dataset by adding 17 problems with manually translated concepts (i.e., with no use of GPT-4o), for which images were also selected manually, and 7 constructed by hand, i.e., by means of making photos of manually-built scenes reflecting the respective concepts.

All in all, out of 60 real-world problems obtained (Bongard-RWR), 12 were generated fully automatically, 24 were generated automatically and then adjusted manually (manual selection of the images), and 24 were constructed entirely manually. Appendix F presents the details.

Dataset variants. Bongard-RWR was designed to capture real-world complexity, including variations in aspect ratio, resolution, and layout. To complement the base dataset, we developed three additional variants: RWR-S (square images), RWR-G (greyscale images), and RWR-SG (square greyscale images). In the square image variants, we manually cropped all images to square format and rescaled them to 512×512 resolution. The BP matrix was resized to 1024×1024 width while maintaining aspect ratio, resulting in a height of 763px.

5. Experiments

To evaluate the AVR capabilities of MLLMs, we conduct experiments in two main settings, involving 3 binary clas-

sification setups and 7 proposed generation methods. Our evaluation spans a range of MLLMs, including 4 proprietary models accessible via API: GPT-4o, GPT-4 Turbo (Achiam et al., 2023), Gemini 1.5 Pro (Reid et al., 2024), and Claude 3.5 Sonnet (Anthropic, 2024), alongside 4 open-access models run locally on an NVIDIA DGX A100 node: InternVL2-8B (Chen et al., 2024c;b), LLaVA-1.6 Mistral-7B (Liu et al., 2023; 2024; Jiang et al., 2023), Phi-3.5-Vision (Abdin et al., 2024), and Pixtral 12B (MistralAI, 2024). We consider four BP datasets covering both synthetic and real-world images. To balance diversity and computational feasibility, we cap the dataset size at 100 instances. Specifically, we use the first 100 manually constructed (synthetic) BPs from (Bongard, 1970), 100 problem samples from each of the Bongard HOI and Bongard-OpenWorld datasets selected from their test splits through stratified sampling based on problem concept labels, and all 60 instances from Bongard-RWR. Extended results are presented in Appendix C.

Binary classification. Fig. 4 presents the results of binary classification tasks. In the Ground-Truth setting, most proprietary and some open-access models outperform a random classifier baseline. In the Incorrect Label setting, since rejecting incorrect concepts is a generally easier task, most models perform better than in the Ground-Truth setup. However, the consistently shifted performance of open-access models suggests a potential bias toward agreeing or disagreeing with provided concepts. In the Images to Sides task, proprietary models demonstrate strong performance, while